ELIF S. BESIKTEPE

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GitHub: ElifSB-creator — HuggingFace — LinkedIn

RESEARCH SUMMARY

Junior AI Researcher focused on Scientific Machine Learning and Computational Astrophysics

Researcher pivoting a quantitative background in Economics toward Artificial Intelligence (since 2025) and Computational Astrophysics (since 2026). Applying statistical data analysis expertise to exoplanet datasets from Kepler and TESS through Bayesian inference, MCMC-based scientific modeling, and uncertainty quantification processes. Focused on extracting physical insights from complex datasets and developing interpretable AI solutions within the Scientific Machine Learning framework.

TECHNICAL COMPETENCIES

- **Mathematical Foundation:** Probability Theory, Bayesian Statistics, Regression Analysis, Econometric Time Series Analysis, and Forecasting.
- **AI/ML & Data Science:** Deep Learning (Hugging Face Transformers), Model Evaluation, NLP, Vision-Language Models (LLaMA-3), MLOps, Streamlit, K-Means Clustering.
- **Astronomy Research:** NASA Exoplanet Archive, `Lightkurve`, `Astropy`, `BATMAN`, Gaia DR3 Database.
- **Programming & DevOps:** Python (Scientific Stack: NumPy, Pandas, Matplotlib, SciPy), LaTeX, Git, GitHub Terminal Workflow.

SELECTED RESEARCH AND ENGINEERING PROJECTS

Transit Photometry Parameter Estimation Pipeline *Computational Astrophysics Research (2026)*

- **Problem:** Quantitative analysis of systematic uncertainties and biases in planetary radius (R_p/R_s) estimations caused by fixed "limb darkening" assumptions in transit photometry.
- **Action:** Modeled HD 209458b, Kepler-186f, and TOI-700 d systems using the **Mandel & Agol (2002)** analytical framework. Applied Bayesian inference via **MCMC (Markov Chain Monte Carlo)** sampling for orbital parameter extraction and uncertainty quantification.
- **Result:** Quantified through simulation experiments that allowing limb darkening coefficients to float reduces systematic overestimation bias in radius by **5.1% ± 1.2%**.

Intelligent OCR & Document Insight System (LLaMA Vision) *AI Engineering Project (2026)*

- **Problem:** The need for structured information extraction from complex academic and technical documents while preserving context and technical meaning.
- **Action:** Integrated the `meta-llama/llama-3.2-11b-vision-instruct` model into a Streamlit-based interface. Optimized **low temperature (0.1)** settings and prompt engineering techniques to minimize hallucination risks.
- **Impact:** Developed an end-to-end AI prototype with high-accuracy text extraction, translation, and summarization capabilities for scientific publications and technical reports, deployed on Hugging Face.
- **Link:** HuggingFace Demo

RESEARCH TRAINING

Intro2Astro Astronomy Research Workshop August 2026
Researchers affiliated with UC Berkeley & University of Hawaii Online

- **Scope:** Intensive research training on exoplanet detection, characterization, and Bayesian statistical modeling using Python-based scientific workflows.
- **Methodology:** Light curve analysis (detrending) on transit photometry data, orbital parameter estimation, and **MCMC sampling** for uncertainty quantification applications.
- **Tools Used:** Effective utilization of industry-standard astronomy libraries including `Lightkurve`, `Astropy`, and `BATMAN`, along with **NASA Exoplanet Archive** and **Gaia DR3** data sources.

TECHNICAL SKILLS

- **Programming:** Python, NumPy, Pandas, Matplotlib, SciPy, LaTeX, Git/GitHub

- **Machine Learning:** Machine Learning, Deep Learning, NLP, Computer Vision, Vision-Language Models, Model Evaluation
- **Data Science:** Statistical Analysis, Regression, Time Series Analysis, Data Visualization
- **Scientific Computing:** Bayesian Statistics, MCMC Sampling, Uncertainty Quantification, Scientific Machine Learning
- **Astronomy Tools:** Astropy, Lightkurve, BATMAN, NASA Exoplanet Archive, Gaia DR3
- **Deployment Tools:** Streamlit, Hugging Face Spaces

EDUCATION

Bachelor of Science in Economics (BSc) *Focused on Statistics and Econometrics* 2015

- Academic foundation in statistics, econometrics, quantitative analysis, data interpretation, and mathematical modeling.

CERTIFICATIONS

- Overleaf Novice Learning Path Certificate (2026)

RESEARCH INTERESTS

Scientific Machine Learning, Bayesian Inference, Computational Astrophysics, Exoplanet Characterization, AI-assisted Scientific Discovery